Detecting AI-Driven Fraudulent Amazon Reviews Using Big Data Similarity Analysis

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01

Executive Summary

Steps and Observations:

- Processed 64M+ Amazon reviews and 4.3M metadata records; retained 63.9M clean reviews
- Review activity shows strong growth (2013–2019), a 2020 spike, and high but volatile volume through 2023
- Seasonal patterns: holiday peaks, March uptick, and Prime Day spikes across categories
- High-review products cluster around 4.0–4.6 stars; extreme ratings tend to have fewer reviews
- Top reviewers span many different product categories
- Titles duplicate far more than full texts; text reviews remain highly unique across distance thresholds
- Duplication increases as Jaccard distance threshold rises (0.3 → 0.7)
- Recent reviews are more unique, suggesting strong modern ML-based moderation fraud detection

Recommendation:

- Use a MinHash + Jaccard pipeline to detect fraudulent Amazon reviews
- Use title similarity with low distance thresholds (0.3) to detect mass-posted or repetitive content
- Apply low thresholds on full texts only to flag very high-similarity cases and avoid false positive

02

Data Overview

Reviews Data

Column	Type
asin	string
helpful_vote	long
parent_asin	string
rating	double
text	string
timestamp	long
title	string
user_id	string
verified_purchase	boolean

- Source: gs://msca-bdp-data-open/final_project_reviews/reviews_parquet
- Count: 64,679,785
- Detailed description about each review

Meta Data

Column	Type
author	struct
average_rating	double
bought_together	string
categories	array
description	array
main_category	string
parent_asin	string
price	string
rating_number	long
store	string
subtitle	string
title	string

- Source: gs://msca-bdp-data-open/final_project_reviews/meta_parquet
- Count: 4,320,533
- Product-level attributes and metadata

03

Data Cleaning

Reviews Data

- No null values
- Removed 711, 229 duplicates
- Converted timestamps → human-readable dates
- Filtered out:
  - Null parent_asin, asin
  - Ratings outside 1-5
  - helpful_votes < 0
  - Null verified_purchase
  - Future timestamps
- Kept columns:
  - parent_asin
  - rating
  - title
  - text
  - user_id
  - review_date
- Final Count: 63,968,550

Meta Data

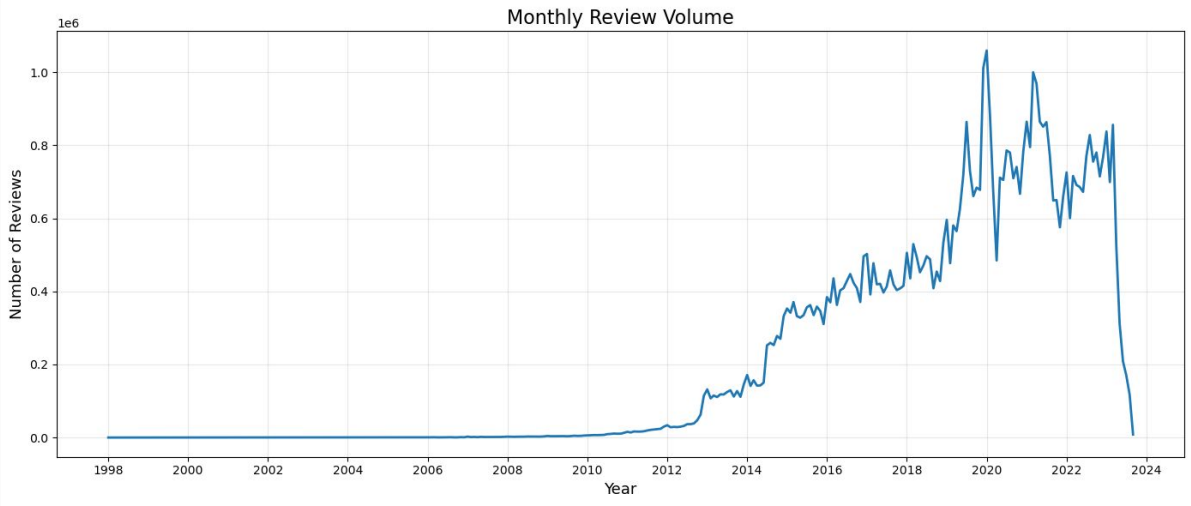
- Cleaned whitespace in main_category
- 99% null values for author, bought_together, subtitle
- No Duplicates
- Converted price from string → float
- Filtered out:
  - Ratings outside 1-5
  - Null categories, main_category
  - Null parent_asin
  - price < 0
  - rating_number < 1
- Kept columns:
  - parent_asin
  - average_rating
  - main_category
  - price
- Final Count: 1,714,238

04

Timeline of Review Activity

Observations:

- Earliest Review: 1998-01-19
- Latest Review: 2023-09-13
- Early years show sparse data collection
- Left skewed — more recent reviews
- Collection gaps in months:
  - May 1998
  - August 1998
  - October 1998
  - November 1998
  - March 1999
  - April 1999
  - May 1999
  - September 1999
- 1 IQR outlier
  - January 2020
  - Possibly early COVID-related, though slightly ahead of global surge



05

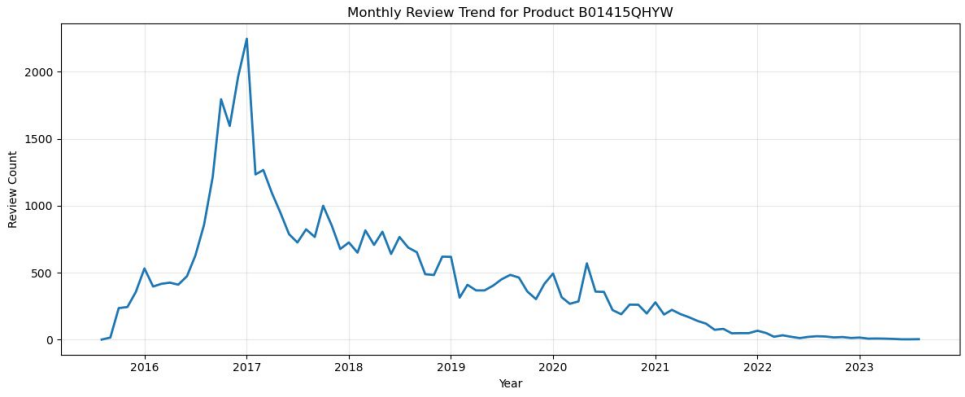
Top 10 Most Reviewed Products

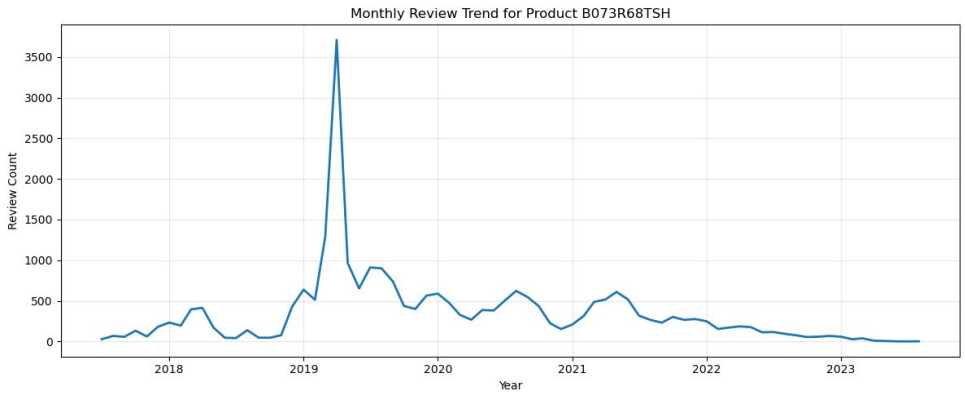
Top 10 by Review Count:

Product Asin	Total Review Count
B01415QHYW	42771
B01LSUQSB0	36823
B0BVGHXZJ1	33835
B07P9S6R9Y	29029
B001MA0QY2	27299
B0C7CZXBBT	25531
B073R68TSH	25390
B01ITNRGFG	25218
B07H2V5YLH	24529
B0BYBCFHTB	24136

Trend Observations:

- Review volume starts at 0 and has an exponential acceleration within first 2 years
- Generally has one peak at around 1k–3k reviews/month



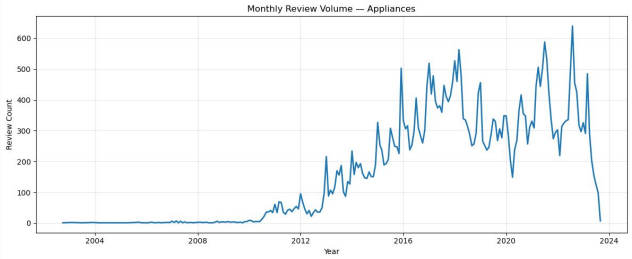


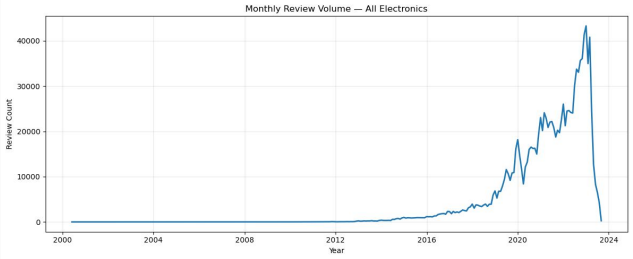
06

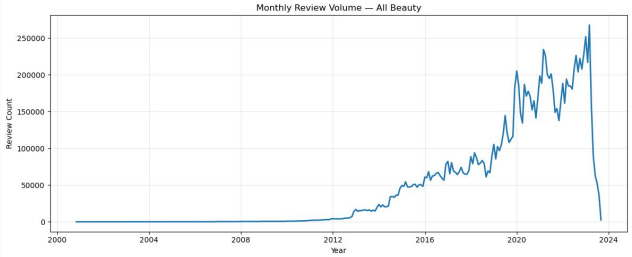
Review Volume by Product Category – Overall

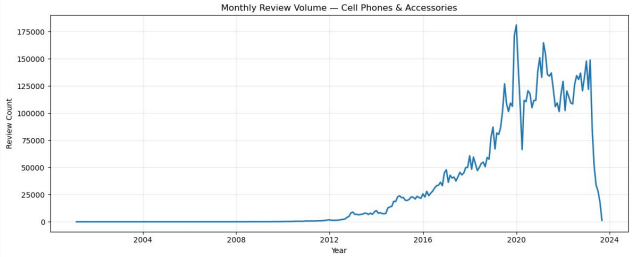
Overall Observations:

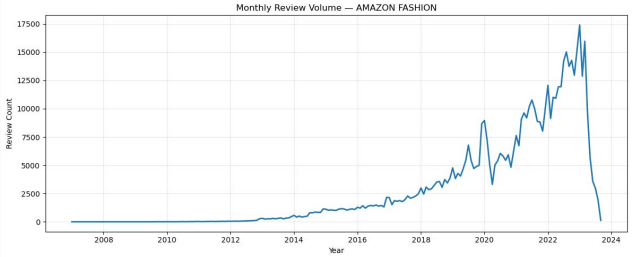
- Most graphs are left skewed
  - Minimal activity before 2013
  - Strong growth 2013–2019
  - Pandemic spike in 2020
- Elevated but volatile during 2021–2023

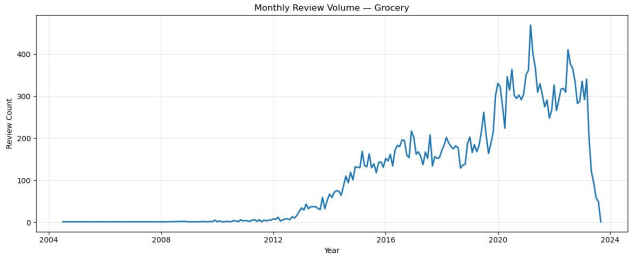










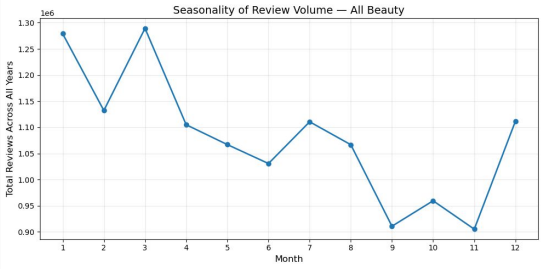


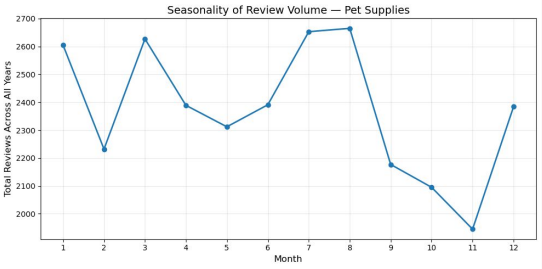
07

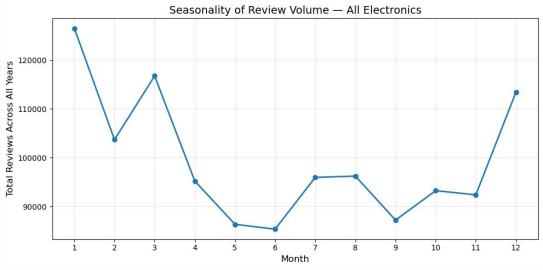
Review Volume by Product Category – Seasonality

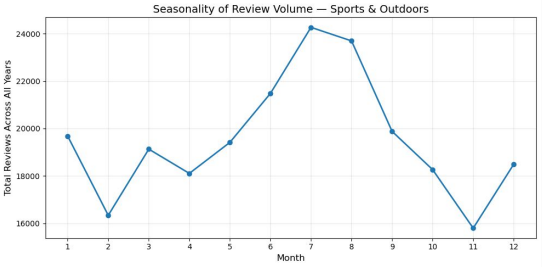
Seasonality Observations:

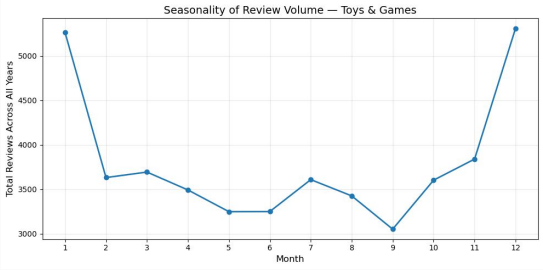
- Clear holiday-season peak in December–January
- Secondary March uptick, possibly due to post-holiday reviews or seasonal promotions
- Prime Day spike in July
- Patterns vary slightly by product category
    - Sports & Outdoors show strong mid-year activity, likely driven by warmer weather

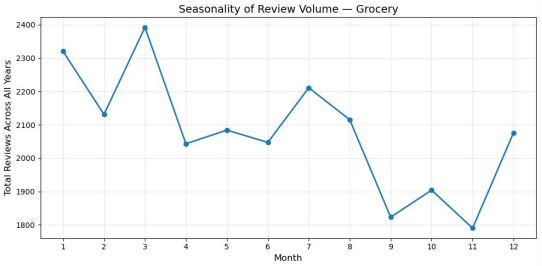










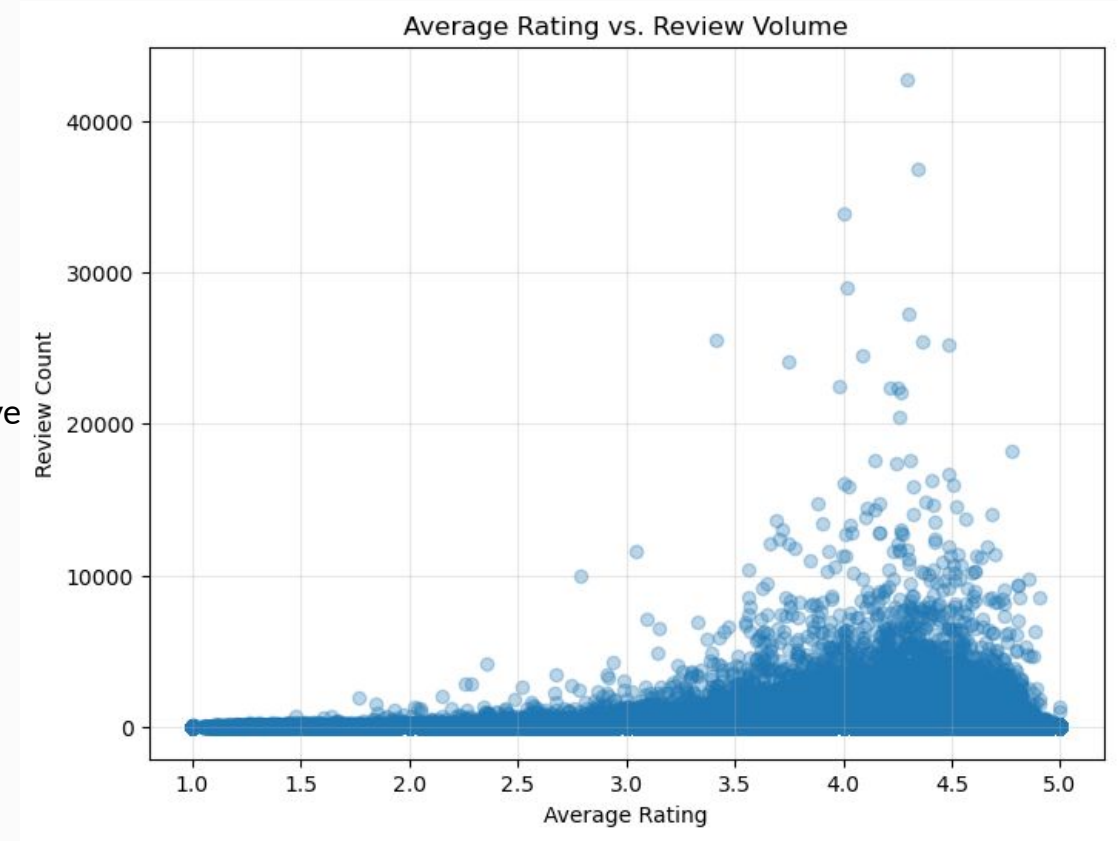


08

Average Ratings vs. Review Counts

Observations:

- Rating distribution is left-skewed
- Highest-rated products (4.9–5.0) typically have few reviews
- Lowest-rated products typically have few reviews
    - Low ratings deter purchases and therefore reviews
- Most-reviewed products cluster around the 4.0–4.6 range

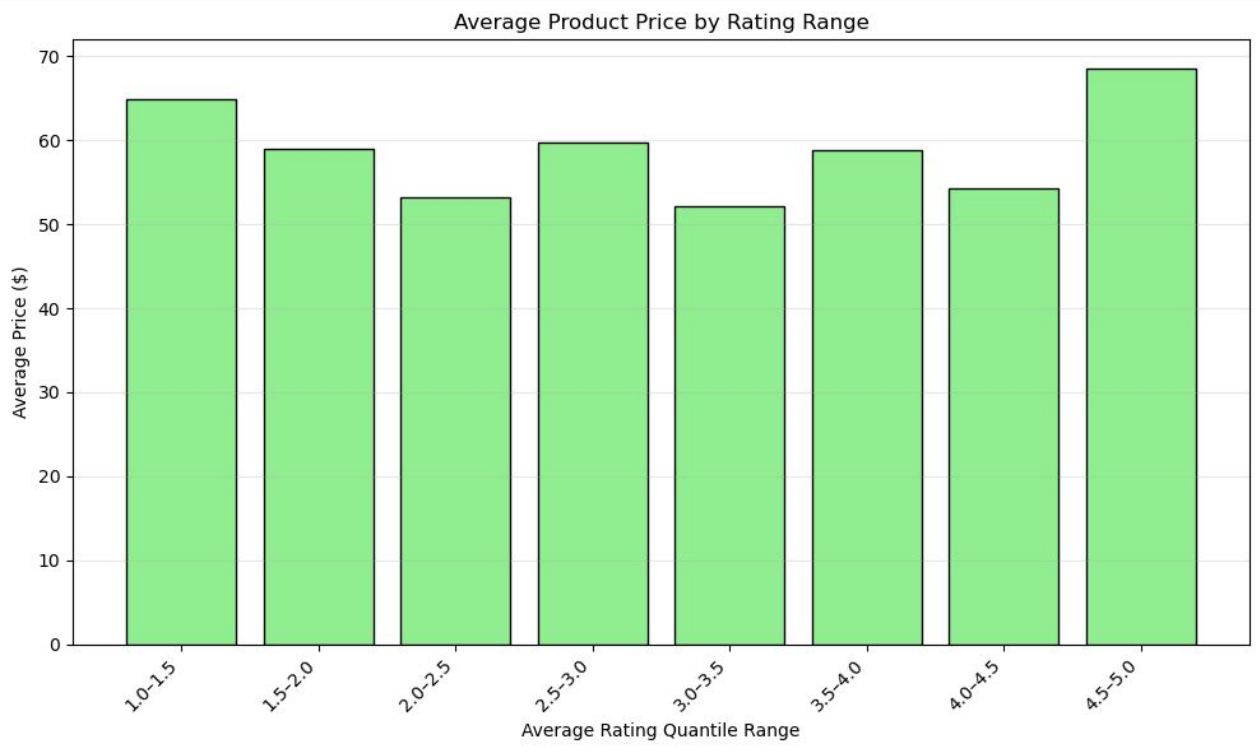


09

Average Rating vs. Average Product Price

Observations:

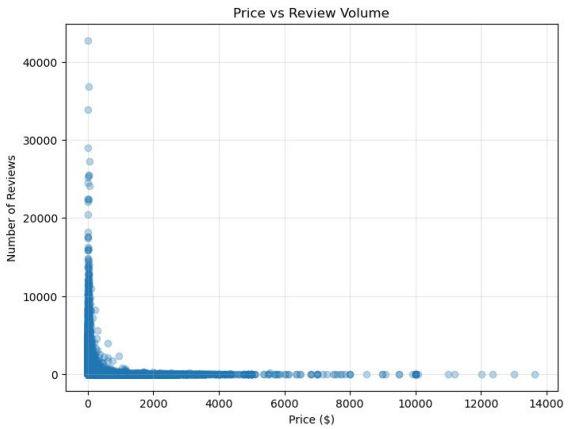
- The lowest and highest rated products show slightly higher average prices
  - Premium products tend to earn very high ratings
  - High-priced but disappointing products may accumulate very low ratings
- Generally uniform distribution

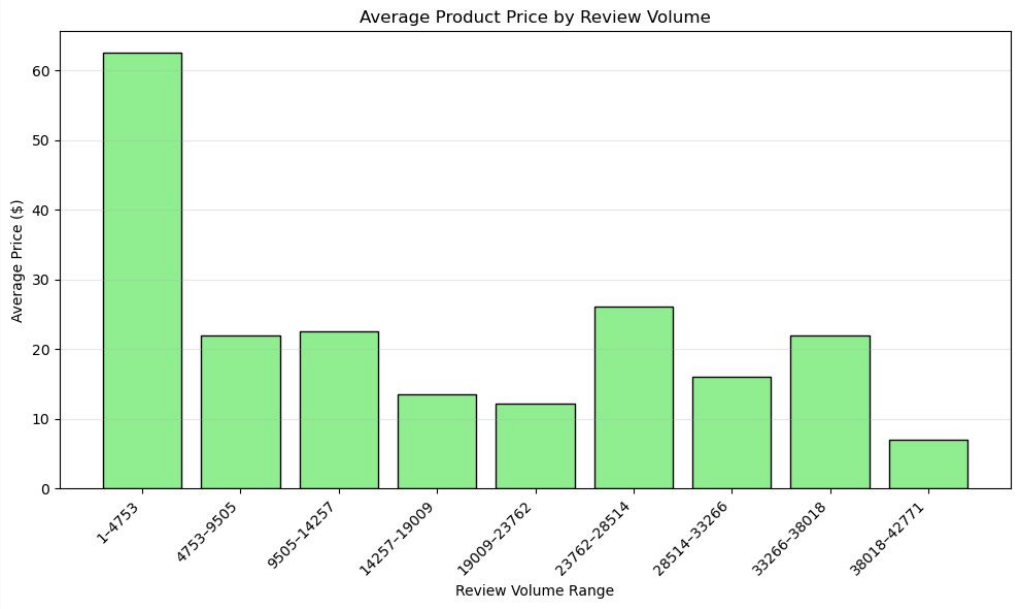


Review Volume vs. Product Price

Observations:

- Right skewed
- Lower review volume products tend to have higher average prices
  - Fewer people buy expensive products
- At higher review volumes, price levels converge around $10-25





Common Words in Reviews and Titles

5 ⭐ Reviews

Title		Text	
Word	Count	Word	Count
great	7383685	great	11235305
stars	5064531	love	6978520
five	5023701	product	6528079
good	3210245	good	5992631
product	2795945	like	5214722
love	2405655	easy	5167377
works	1898809	phone	5164077
perfect	1721789	use	4994016
case	1369068	one	4776125
easy	1307726	case	4721045

1 ⭐ Reviews

Title		Text	
Word	Count	Word	Count
one	628256	product	1548204
star	518803	one	1357315
dont	474052	phone	1322091
buy	405886	like	1106560
work	363219	case	925364
money	333380	dont	917775
fit	333153	work	909654
product	329500	use	853990
doesnt	299244	get	839548
waste	267076	money	804380

Observations:

- Positive words cluster near 5-star reviews; negative words near 1-star
- Common product terms appear across all ratings (e.g., *case*, *phone*)

Most Active Reviewers

Observations:

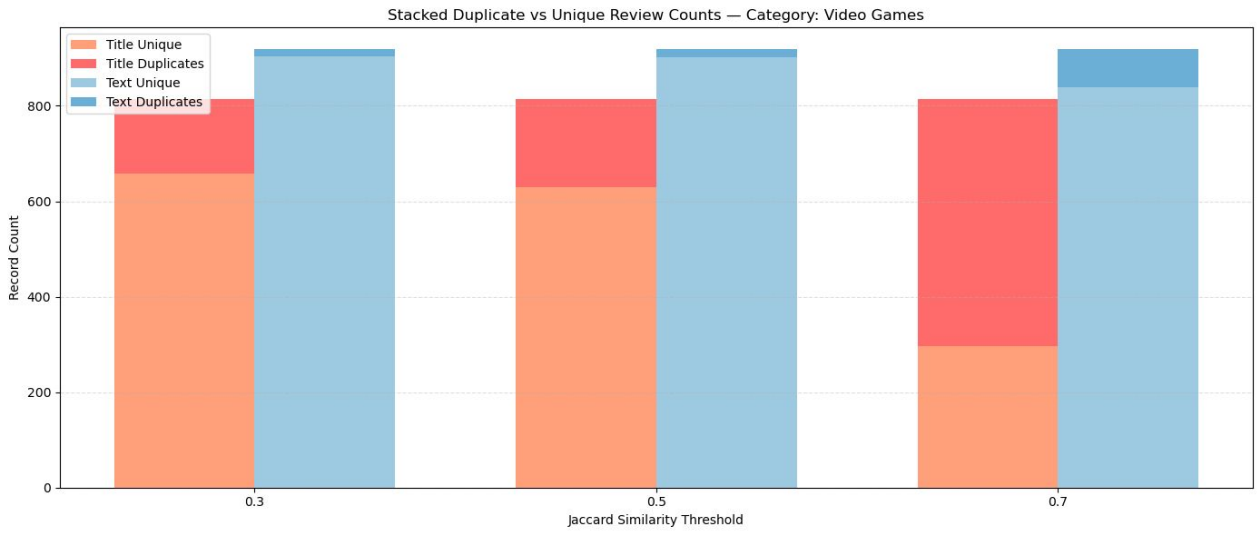
- Top 10 users post 1.3k–3.4k reviews across 12–21 categories
- Very high cross-category reviewing activity
- The top reviewer is an outlier, spanning 21 categories as opposed to others in the 12-15 range

User ID	Total Reviews	Number of Categories
AG73BVBKUOH22USSFJA5ZWL7AKXA	3448	21
AGZUJTI7A3JFKB4FP5JOH6NVAJIQ_1	2394	13
AEMP3A7IKW37CMWFXNKXWW6HGJHA_1	2328	12
AFXF3EGQTQDXMRLDWFU7UBFQZB7Q	2031	14
AH2DN3Q2OKAW7JAFRIYPHOUOM2ZQ	1917	13
AEAXAJACFMXIAAH4WOHRMXPSZWFA	1849	14
AEZP6Z2C5AVQDZAJECQYZWQRNG3Q	1784	12
AFF6DS6NOWRIF2AROFASUVVOPYLA	1574	14
AHV6QCNBJNSGLATP56JAWJ3C4G2A	1480	15
AGYVC7KVHP2AWM7BDCEYNHFA6F3Q	1358	14

Similarity Analysis (Video Games Category, 10% Sample)

Observations:

- Title duplication rises sharply as the Jaccard distance threshold increases
- Text reviews remain mostly unique but also follows similar pattern
- Short titles capture duplication far more easily than full review text



Similarity Analysis Across Top 5 Products in Video Game Category – Part 1

Title

Product	Total Titles	Dup @0.3	Dup @0.5	Dup @0.7
B08FYC9DQL	168	18	21	88
B07RFTRFPZ	192	32	39	110
B0BZ8DMFMM	58	9	11	32
B0851CKX2H	24	5	5	14
B0BTX52WS7	4	0	0	1

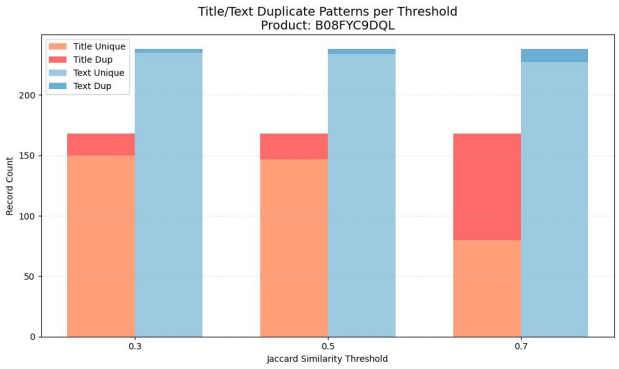
Text

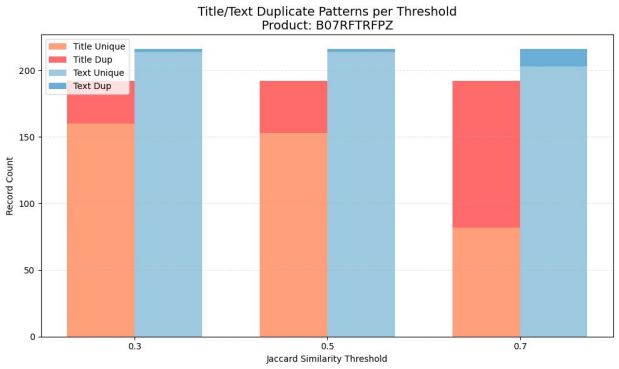
Product	Total Texts	Dup @0.3	Dup @0.5	Dup @0.7
B08FYC9DQL	238	3	4	11
B07RFTRFPZ	216	2	2	13
B0BZ8DMFMM	63	0	0	7
B0851CKX2H	40	0	1	4
B0BTX52WS7	14	0	0	0

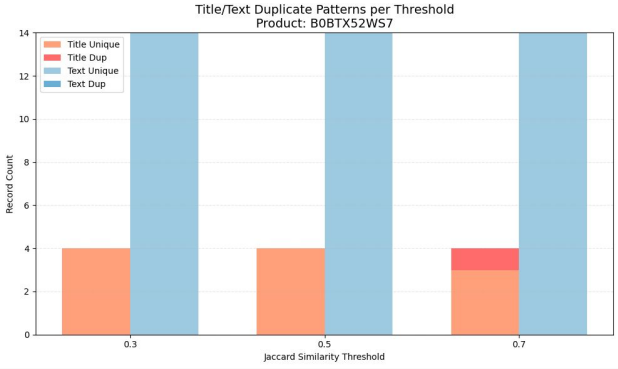
Similarity Analysis Across Top 5 Products in Video Game Category – Part 2

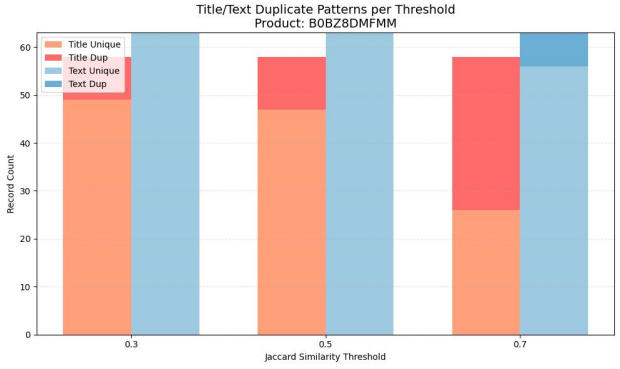
Observations:

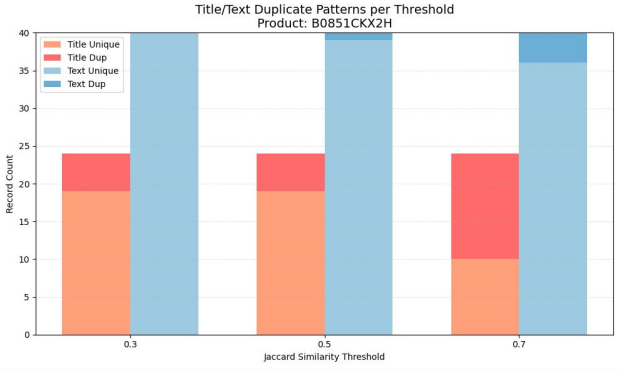
- Follows the same overall pattern across products
- Some items have too few reviews, resulting in zero duplicates
- Title duplication is far more common than text duplication
- Duplication rises sharply as the distance threshold increases from 0.3 → 0.7







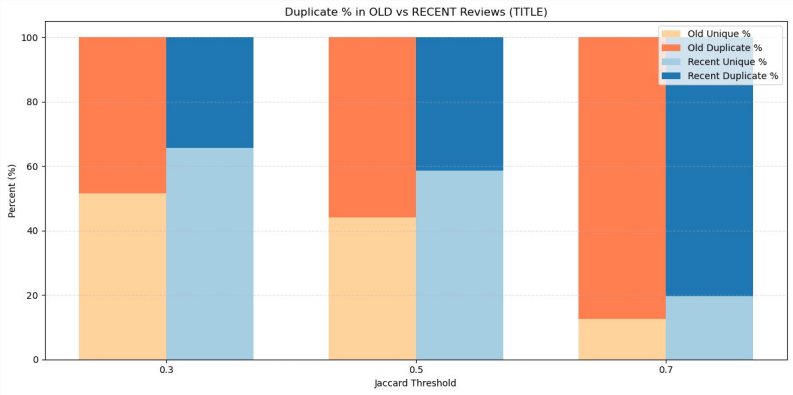


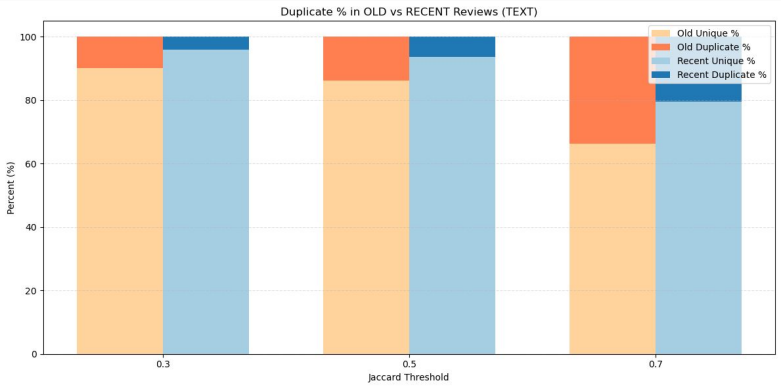


Similarity Detection on Recent vs. Older Reviews (0.1% sample)

Observations:

- Titles duplicate far more than full review texts
- Recent reviews are more unique than older ones
- One might expect more bots generating repetitive content today, but modern ML-based detection systems likely suppress many of these duplicates





Methodology for All Similarity Analysis

- Filter by group and sample if necessary
- Performed MinHash + Jaccard similarity to detect near-duplicate titles and full texts
- Evaluated duplication at three distance thresholds: 0.3, 0.5, 0.7
    a. Note: a threshold of *0.3* is the distance and means similarity ≥ 0.7 is required to count as a duplicate
- Computed per-group duplicate vs. unique counts and percentage

Conclusion and Actionable Recommendation

Current State / Overall Trends:

- Short titles create high similarity from small overlaps → easier duplicate detection
- Full review texts remain largely unique
- Duplicate rates have declined in recent years, suggesting that current detection systems are increasingly effective

Recommendation:

- Use a MinHash + Jaccard similarity system to detect near-duplicate titles and full texts
- For detecting mass-posted or suspicious reviews, rely primarily on title-based similarity, but using a lower distance threshold (0.3) since short titles naturally produce higher similarity scores
- For stricter fraud detection, use full text reviews and apply lower distance thresholds (e.g., 0.3) to require very high similarity, ensuring that only clearly suspicious reviews are flagged and minimizing false positives

Thank You

If something comes up later, contact amizuno@uchicago.edu